Exp 14 11/10/24	Write a code using RAW Sockets to implement packet sniffing

Aim:

To write a code using RAW sockets to implement packet sniffing.

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown"

        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("-" * 50)

sniff(iface='Wi-Fi', prn=packet_callback)
```

Output:

```
Protocol: TCP
Source IP: 57.155.144.117
Destination IP: 192.168.1.6

Protocol: TCP
Source IP: 192.168.1.6
Destination IP: 57.155.141.197

Protocol: UDP
Source IP: 192.168.1.6
Destination IP: 192.168.1.1

Protocol: UDP
Source IP: 192.168.
Destination IP: 192.168.1.1
```

Result

Thus the program was implemented using RAW Sockets to implement packet sniffing